AFRILANG Stdlib

std/c/simmarkov

Français. Bibliothèque AFRILANG ‘std/c/simmarkov‘ (niveau complex, catégorie complex). Elle expose 6 fonction(s) : steadyState, simulate, transitionProb, expectedSteps, isStochastic, mixingTime.
‘import ”std/c/simmarkov”‘ · ‘use simmarkov‘
- ‘steadyState(trans list number, states number) → list number‘ - ‘simulate(trans list number, states number, steps number, seed number) → list number‘ - ‘transitionProb(trans list number, states number, from number, to number) → number‘ - ‘expectedSteps(trans list number, states number, target number) → number‘ - ‘isStochastic(trans list number, states number) → bo **English.** AFRILANG library ‘std/c/simmarkov‘ (tier complex, category complex). Exports 6 function(s): steadyState, simulate, transitionProb, expectedSteps, isStochastic, mixingTime.
‘import ”std/c/simmarkov”‘ · ‘use simmarkov‘
- ‘steadyState(trans list number, states number) → list number‘ - ‘simulate(trans list number, states number, steps number, seed number) → list number‘ - ‘transitionProb(trans list number, states number, from number, to number) → number‘ - ‘expectedSteps(trans list number, states number, target number) → number‘ - ‘isStochastic(trans list number, states number) → bool‘ - ‘mixing

Catégorie / Category	Modules complex (c/)
Niveau / Tier	complex
Fonctions / Functions	6

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Contents

Français

1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/simmarkov' (niveau complex, catégorie complex). Elle expose 6 fonction(s) : steadyState, simulate, transitionProb, expectedSteps, isStochastic, mixingTime.

'import "std/c/simmarkov"' · 'use simmarkov'

```
- `steadyState(trans list number, states number) → list number`
- `simulate(trans list number, states number, steps number, seed number) → list
   number`
- `transitionProb(trans list number, states number, from number, to number) → number`
- `expectedSteps(trans list number, states number, target number) → number`
- `isStochastic(trans list number, states number) → bool`
- `mixingTime(trans list number, states number) → number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/simmarkov"
use simmarkov
```

```
Niveau : complex · Catégorie : Modules complex (c/)
Domaines avancés — graphes, NLP, réseaux complexes.
```

3. Référence API

```
•
 steadyState(trans list number, states number) → list•
 simulate(trans list number, states number, steps number, seed number) → list•
 transitionProb(trans list number, states number, from number, to number) → number•
 expectedSteps(trans list number, states number, target number) → number•
 isStochastic(trans list number, states number) → bool•
 mixingTime(trans list number, states number) → number
```

4. Exemples d'utilisation

```
import "std/c/simmarkov"
use simmarkov
```

```
create result = steadyState(/* args */)
say result
```

```
create result = simulate(/* args */)
say result
```

```
create result = transitionProb(/* args */)
say result
```

```
create result = expectedSteps(/* args */)
say result
```

```
create result = isStochastic(/* args */)
say result
```

```
create result = mixingTime(/* args */)
say result
```

5. Bonnes pratiques

```
•
 Préférez `import "std/c/simmarkov"` en tête de fichier.•
 Lancez `afrilang check` avant de livrer.•
 Combinez avec les modules stdlib de la même catégorie.•
 Voir /stdlib/ pour le source live et les démos playground.•
 Signalez les problèmes sur GitHub si une export manque.
```

6. Annexe — source

module simmarkov

```
function steadyState(trans list number, states number) returns list number
end
```

```
function simulate(trans list number, states number, steps number, seed number)
    returns list number
end
```

```
function transitionProb(trans list number, states number, from number, to number)
    returns number
end
```

```
function expectedSteps(trans list number, states number, target number) returns
    number
end
```

```
function isStochastic(trans list number, states number) returns bool
end
```

```
function mixingTime(trans list number, states number) returns number
end
end
```

English

1. Overview

AFRILANG library 'std/c/simmarkov' (tier complex, category complex). Exports 6 function(s): steadyState, simulate, transitionProb, expectedSteps, isStochastic, mixingTime.

'import "std/c/simmarkov"' · 'use simmarkov'

```
- `steadyState(trans list number, states number) → list number`
- `simulate(trans list number, states number, steps number, seed number) → list
   number`
- `transitionProb(trans list number, states number, from number, to number) → number`
- `expectedSteps(trans list number, states number, target number) → number`
- `isStochastic(trans list number, states number) → bool`
- `mixingTime(trans list number, states number) → number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/simmarkov"
use simmarkov
```

```
Tier: complex · Category: Complex modules (c/)
Advanced domains — graphs, NLP, complex networks.
```

3. API reference

```
•
 steadyState(trans list number, states number) → list•
 simulate(trans list number, states number, steps number, seed number) → list•
 transitionProb(trans list number, states number, from number, to number) → number•
 expectedSteps(trans list number, states number, target number) → number•
 isStochastic(trans list number, states number) → bool•
 mixingTime(trans list number, states number) → number
```

4. Usage examples

```
import "std/c/simmarkov"
use simmarkov
```

```
create result = steadyState(/* args */)
say result
```

```
create result = simulate(/* args */)
say result
```

```
create result = transitionProb(/* args */)
say result
```

```
create result = expectedSteps(/* args */)
say result
```

```
create result = isStochastic(/* args */)
say result
```

```
create result = mixingTime(/* args */)
say result
```

5. Best practices

```
•
 Prefer `import "std/c/simmarkov"` at the top of your file.•
 Run `afrilang check` before shipping.•
 Combine with related stdlib modules from the same category.•
 See /stdlib/ for the live source and playground demos.•
 Report issues on GitHub if an export is missing or undocumented.
```

6. Appendix — source

module simmarkov

```
function steadyState(trans list number, states number) returns list number
end
```

```
function simulate(trans list number, states number, steps number, seed number)
    returns list number
end
```

```
function transitionProb(trans list number, states number, from number, to number)
    returns number
end
```

```
function expectedSteps(trans list number, states number, target number) returns
    number
end
```

```
function isStochastic(trans list number, states number) returns bool
end
```

```
function mixingTime(trans list number, states number) returns number
end
end
```

Référence rapide / Quick reference

- Import : `import "std/c/simmarkov"`
- Vérification : `afrilang check`
- Documentation : `https://afrilang.dev/stdlib/std/c/simmarkov/`

Source (extrait)

```
module simmarkov

function steadyState(trans list number, states number) returns list number
end

function simulate(trans list number, states number, steps number, seed number)
    returns list number
```

```
end

function transitionProb(trans list number, states number, from number, to number)
    returns number
end

function expectedSteps(trans list number, states number, target number) returns
    number
end

function isStochastic(trans list number, states number) returns bool
end

function mixingTime(trans list number, states number) returns number
end
end
```